

Foundation (Jalapeno)

- Q1.** Write the quadratic function $f(x) = x^2 + 4x + 4$ in vertex form.
- (A) $f(x) = (x + 2)^2$
 (B) $f(x) = (x - 2)^2$
 (C) $f(x) = x^2 + 4x + 4$
 (D) $f(x) = (x + 4)^2$
 (E) $f(x) = (x - 4)^2$
- Q2.** The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$. What does h represent?
- (A) the y -intercept
 (B) the x -intercept
 (C) the vertex's x -coordinate
 (D) the vertex's y -coordinate
 (E) the axis of symmetry
- Q3.** Write the quadratic function $f(x) = (x - 2)(x + 3)$ in intercept form.
- (A) $f(x) = x^2 + x - 6$
 (B) $f(x) = x^2 + x + 6$
 (C) $f(x) = (x - 2)(x + 3)$
 (D) $f(x) = x^2 - x - 6$
 (E) $f(x) = x^2 - x + 6$
- Q4.** The quadratic function $f(x) = ax^2 + bx + c$ has x -intercepts at $(2, 0)$ and $(-3, 0)$. What is the factored form of $f(x)$?
- (A) $f(x) = (x - 2)(x + 3)$
 (B) $f(x) = (x + 2)(x - 3)$
 (C) $f(x) = a(x - 2)(x + 3)$
 (D) $f(x) = a(x + 2)(x - 3)$
 (E) $f(x) = (x - 2)^2(x + 3)$
- Q5.** The quadratic function $f(x) = x^2 - 4x - 5$ can be written in vertex form as $f(x) = a(x - h)^2 + k$. What is the value of a ?
- (A) 1
 (B) -1
 (C) 2
 (D) -2
 (E) 5
- Q6.** The quadratic function $f(x) = (x + 1)^2 - 4$ has a vertex at (h, k) . What is the value of k ?
- (A) -4
 (B) -1
 (C) 1
 (D) 2
 (E) 4
- Q7.** The quadratic function $f(x) = (x - 2)^2 + 1$ has a y -intercept at $(0, k)$. What is the value of k ?
- (A) 5
 (B) 2
 (C) 1
 (D) 0
 (E) -1
- Q8.** The quadratic function $f(x) = (x + 2)(x - 1)$ can be written in vertex form as $f(x) = a(x - h)^2 + k$. What is the value of h ?
- (A) -1
 (B) 1
 (C) 2
 (D) 3
 (E) 4

Open Ended Questions (FRQ)

- Q9.** Write the quadratic function $f(x) = x^2 + 2x - 6$ in both vertex form and intercept form. Show all work and explain your reasoning.
- Q10.** The quadratic function $f(x) = a(x - 1)(x - 3)$ has x -intercepts at $(1, 0)$ and $(3, 0)$. If the vertex is at $(2, -2)$, find the value of a . Show all work and explain your reasoning.

Chef's Tips

- Emphasize the importance of showing work and explaining reasoning for free-response questions.
- Encourage students to use graphing calculators or graphing software to visualize the quadratic functions.
- Remind students that the vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex.